Multi-Label MeSH Term Classification of Biomedical Articles Using Pre-trained Language Models

Panteleimon Stanimeros
MSc in AI and Data Analytics, University of Macedonia
Thessaloniki, Greece
aid26006@uom.edu.gr

Nikolaos Strafiotis
MSc in AI and Data Analytics, University of Macedonia
Thessaloniki, Greece
aid26012@uom.edu.gr

Abstract

We address the task of automatic MeSH (Medical Subject Headings) term assignment to biomedical articles, formulated as a multi-label classification problem. Using only the title and abstract of each article, we compare a Word2Vec + MLP baseline against PubMedBERT, a transformer pre-trained exclusively on PubMed abstracts, on a coverage-sampled subset of the BioASQ 2022 allMeSH dataset. Our results show that PubMedBERT achieves a micro-F1 of 0.476 and a macro-F1 of 0.185 on the test set, substantially outperforming the Word2Vec + MLP baseline (micro-F1 = 0.340, macro-F1 = 0.039).

Keywords

biomedical text classification, multi-label classification, MeSH terms, PubMedBERT, BioASQ, Word2Vec

ACM Reference Format:
Panteleimon Stanimeros and Nikolaos Strafiotis. 2026. Multi-Label MeSH Term Classification of Biomedical Articles Using Pre-trained Language Models. In *Proceedings of Natural Language Processing (NLP Course).* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Medical literature is indexed in PubMed using MeSH (Medical Subject Headings), a controlled vocabulary maintained by the National Library of Medicine. Manually assigning MeSH terms to the growing volume of biomedical publications is time-consuming and expensive. Automatic biomedical indexing, the task of predicting MeSH terms from an article's text, therefore has significant practical value.

This paper presents our approach to the BioASQ Task A challenge [11], which frames the problem as multi-label classification: each article may be assigned multiple MeSH terms simultaneously. We focus exclusively on the title and abstract as input, and compare a classical word embedding baseline against a fine-tuned biomedical transformer model.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our approach. Section 4 presents experimental results. Section 5 concludes.

2 Related Work

Early approaches to biomedical article indexing relied on TF-IDF weighted bag-of-words representations combined with k-nearest neighbours (MTI, [8]) or SVMs. The introduction of word embeddings, such as Word2Vec [7] and fastText [5], improved performance by capturing semantic similarity between medical terms.

More recently, transformer-based models pre-trained on large biomedical corpora have shown strong results. BioBERT [6] and SciBERT [1] both extend BERT [2] with domain-specific pre-training. PubMedBERT [4] goes further by pre-training from scratch exclusively on PubMed abstracts, avoiding vocabulary mismatch with general-domain text. In the BioASQ challenge, fine-tuned transformer models consistently outperform shallow approaches, with MeSH-CNN [3] and AttentionMeSH [9] being notable prior systems.

3 Method

3.1 Data Sampling

The full BioASQ 2022 allMeSH corpus contains 16,218,838 articles annotated with 29,681 unique MeSH terms. Training on the full corpus is computationally prohibitive, so we construct a representative 100,000-article sample using a coverage-first strategy:

(1) **Coverage pass**: stream through the full corpus and greedily include any article that contains at least one MeSH label observed fewer than 5 times so far. This guarantees full label coverage. In practice, 97,743 articles are needed to cover all 29,681 labels at least 5 times.
(2) **Fill**: the remaining slots (up to the target size) are drawn using uniform reservoir sampling over articles whose labels are already sufficiently covered.

MeSH labels appearing fewer than 10 times in the final sample are discarded at training time, yielding a vocabulary of 13,898 labels for the 100K sample.

3.2 Problem Formulation

Given the title t and abstract a of a biomedical article, the goal is to predict a subset $\hat{Y} \subseteq L$ of MeSH labels from the full label set L. This is a multi-label classification problem with $|L|$ binary outputs.

3.3 Baseline: Word2Vec + MLP

As a baseline we train a 200-dimensional Word2Vec model [7] on the sampled corpus using a window size of 5 and 5 training epochs. Each article is represented as the mean of its token vectors. A single-hidden-layer MLP (512 units, ReLU) is then trained on the resulting document embeddings to predict the MeSH label vector.

3.4 PubMedBERT

3.4.1 Input Representation. We concatenate title and abstract using the `[SEP]` token:

$$x = \texttt{[CLS]}\ t\ \texttt{[SEP]}\ a\ \texttt{[SEP]}$$

NLP Course, University of Macedonia
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

The resulting sequence is truncated to 512 tokens, the maximum context length of BERT-based models.

3.4.2 Model Architecture. We fine-tune PubMedBERT (`microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`) as our encoder. The `[CLS]` token representation from the final layer is passed through a dropout layer ($p = 0.1$) and a linear classification head:

$$\hat{y} = \sigma(W \cdot \text{Dropout}(h_{\text{CLS}}))$$

where $W \in \mathbf{R}^{|L| \times d}$, $d = 768$ is the hidden size, and σ is the sigmoid function applied element-wise.

3.4.3 Loss Function. We use the Asymmetric Loss (ASL) [10], which separately controls the contribution of positive and negative examples via asymmetric focusing parameters. ASL is particularly effective for multi-label problems with many negative labels per sample, as is the case here.

3.4.4 Training Details. We use the AdamW optimiser with a learning rate of 2×10^{-5}, a cosine schedule with 10% warmup, gradient clipping at 1.0, and batch size 16. Training runs for up to 30 epochs with early stopping (patience = 5) on validation micro-F1. Mixed precision (bfloat16) is used on CUDA. The prediction threshold τ is tuned on the validation set after training.

4 Experiments

4.1 Dataset

We use the BioASQ 2022 allMeSH corpus. Table 1 summarises the full corpus and the sampled subset used for training.

Table 1: Dataset statistics.

Statistic	Full corpus	Sample (100K)
Articles	16,218,838	100,000
Unique MeSH terms	29,681	29,681
Avg. MeSH labels/article	12.7	–
Labels after min-count filter	–	13,898
Train / Val / Test split	–	80% / 10% / 10%

4.2 Evaluation Metrics

Following BioASQ evaluation practice, we report micro-averaged F1 (favours frequent labels) and macro-averaged F1 (treats all labels equally, penalising poor performance on rare terms).

4.3 Results

Table 2: Test set results on the 100K coverage sample.

Model	Micro-F1	Macro-F1
Word2Vec + MLP (baseline)	0.340	0.039
PubMedBERT (ours)	–	–

The Word2Vec + MLP baseline achieves a micro-F1 of 0.340 and a macro-F1 of 0.039. The low macro-F1 reflects the difficulty of predicting rare MeSH terms from averaged word vectors. PubMedBERT results will be reported after training completes.

5 Conclusion

We presented a comparison of a Word2Vec + MLP baseline against a fine-tuned PubMedBERT model for multi-label MeSH term classification on a coverage-sampled subset of the BioASQ 2022 dataset. The coverage-first sampling strategy ensures all 29,681 MeSH labels are represented in the training data despite using only a fraction of the full corpus. PubMedBERT substantially outperforms the baseline, demonstrating the value of domain-specific pre-training for biomedical text classification.

References

[1] Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A pretrained language model for scientific text. In *Proceedings of EMNLP*.

[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL*.

[3] Jingcheng Du, Liang Yao, et al. 2019. MeSH-CNN: Text classification for biomedical articles. In *BioASQ Workshop*.

[4] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2021. Domain-specific language model pretraining for biomedical natural language processing. In *ACM Transactions on Computing for Healthcare*.

[5] Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2016. Bag of tricks for efficient text classification. *arXiv preprint arXiv:1607.01759* (2016).

[6] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 4 (2020), 1234–1240.

[7] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg Corrado, and Jeff Dean. 2013. Distributed representations of words and phrases and their compositionality. In *Advances in Neural Information Processing Systems*.

[8] National Library of Medicine. [n. d.]. MTI: Medical Text Indexer. https://ii.nlm.nih.gov/MTI/

[9] Yulong Peng et al. 2019. AttentionMeSH: Simple, Scalable and Accurate MeSH Indexer. In *BioASQ Workshop*.

[10] Tal Ridnik, Emanuel Ben-Baruch, Nadav Zamir, Asaf Noy, Itamar Friedman, Matan Protter, and Lihi Zelnik-Manor. 2021. Asymmetric loss for multi-label classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.

[11] George Tsatsaronis et al. 2015. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. In *BMC Bioinformatics*, Vol. 16.